ADHD from EEG

Reading Attention in Children's Brain Waves

Capstone Project Report

A multi-channel network that classifies children's EEG as ADHD or control, split honestly by subject — 92% accuracy that actually holds up.

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1. Executive Summary

This project classifies 2-second windows of 19-channel children's EEG as coming from the ADHD or the control group of a research study, using a **multi-channel 1-D CNN** (193,000 parameters). Because there are only 121 children but hundreds of thousands of windows, the single most important choice is the data split.

Splitting **by subject** (no child appears in both training and test), the model reaches **91.7% accuracy and 0.965 ROC-AUC on 24 completely unseen children**. Unlike the ECG project, there is no collapse from validation to test — the signal genuinely transfers across people.

2. Introduction & Motivation

EEG-based ADHD studies are a well-known area — and a well-known trap. Much published work splits EEG windows randomly, letting the same child appear in training and test, so the model learns to recognise individuals rather than ADHD. This project deliberately avoids that.

> **ADHD is diagnosed clinically, never from a brief EEG.** This classifies research-group membership — a pattern-recognition exercise, not a diagnosis.

3. Data

The Nasrabadi ADHD/Control dataset (via Kaggle) contains 19-channel EEG at 128 Hz from 61 ADHD and 60 control children (ages 7–12), recorded during a visual attention task. Signals were cut into non-overlapping 2-second (256-sample) windows, per-channel normalised, and split **by subject** — including a subject-level validation set carved from the training children.

Split	Subjects	Windows
Train	83	5,838
Validation	14	948
Test (unseen)	24	1,614

4. Methodology

A four-block multi-channel 1-D CNN (all 19 electrodes fed together, 193k parameters) was trained with class weighting. Crucially, the model is scored at the **subject level**: each subject's window predictions are averaged into one decision, since a per-window number is optimistic when one child contributes many windows.

5. Results

Metric (unseen subjects)	Value
Subject-level accuracy	91.7% (22 of 24)

Subject-level ROC-AUC	0.965
Window-level accuracy	89.0%

The validation and test numbers agree closely — evidence the model reads something real about the groups rather than memorising individuals.

6. Deployment

Exported to ONNX (775 KB) for an in-browser demo that draws all 19 EEG channels as a clinical-style montage and classifies a chosen recording on-device, and to TensorFlow Lite for an Android app ('EEG Explorer'). Parity with PyTorch was verified to ~3e-6.

7. Limitations & Ethical Considerations

- Classifies research-group membership, not a diagnosis of any individual.
- Only 24 test subjects: 91.7% means 22 of 24 — a wide confidence interval.
- One cohort, one task; real-world attention difficulties are far more varied.

8. Future Work

- Band-limit the input to find which frequencies carry the signal.
- Ablate electrodes to locate which scalp regions matter.
- Subject-level cross-validation over all 121 children to tighten the estimate.